Health Connect Clinic Experience Lab

Week 4 – Data Analytics Initial Analysis

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Track: Data Analytics

1. Introduction and Business Problem

HealthConnect Clinic is looking at how data and AI can be used to improve the patient experience and reduce missed appointments. One of the main challenges identified is the number of patients who do not attend their scheduled appointments. This can result in unused appointment slots and make it more difficult for the clinic to plan and manage its services effectively.

The overall project asks:

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

As a Junior Intern Data Analyst, my role is to focus on the appointment data and use it to understand attendance and no-show patterns. Rather than trying to predict individual patient behaviour at this stage, the focus is on understanding what the existing data can tell us about appointment outcomes.

The Week 4 objective is therefore to establish a clear analytical foundation that can be developed further in Week 5 and the later stages of the project.

2. Resources Reviewed

For this initial analysis, I reviewed the resources provided for the HealthConnect project.

HealthConnect Appointment Dataset

File: HealthConnect_Appointment_Data.csv

The dataset contains fictional and anonymised appointment records. It includes information about:

- Patient demographics
- Appointment details
- Booking information

- Previous appointment history
- Previous no-shows
- Reminder information
- Distance to the clinic
- Waiting time
- Appointment outcomes

HealthConnect Data Dictionary

File: HealthConnect_Data_Dictionary.xlsx

I used the data dictionary to understand what each variable represents before deciding which variables would be useful for the analysis. The assignment identifies the data dictionary as one of the main resources for the Data Analytics track.

3. Dataset Overview

The appointment dataset contains:

- **5,000 appointment records**
- **18 variables**
- **1,696 unique patients**
- **5,000 unique appointment IDs**

The variables cover several areas that are relevant to understanding appointment attendance.

Area	Examples of relevant variables
Patient information	Age, age group, gender
Appointment information	Appointment type, date, day, time
Booking information	Booking date, booking lead days
Previous history	Previous appointments, previous no-shows

Area	Examples of relevant variables
Reminders	Reminder sent, reminder channel
Operational factors	Distance to clinic, waiting time
Outcome	Appointment outcome

The dataset is therefore well aligned with the purpose of the Data Analytics track, which is to investigate appointment attendance and no-show patterns. The assignment specifically asks the Data Analytics track to review the dataset structure, assess its quality, identify relevant variables, develop business questions and propose KPIs.

4. Data Quality Assessment

Method Used

The initial data-quality assessment was carried out using Python and the Pandas library.

I used Python because it provides a systematic way to inspect the dataset rather than relying only on manual inspection. The checks focused on the structure, completeness, uniqueness and consistency of the data.

The main checks performed were:

Data-quality check	Method used
Number of records and variables	Checked the dataset shape
Column names	Reviewed the dataset columns
Data types	Inspected the data types of each variable
Missing values	Counted missing/null values for each column
Duplicate records	Checked for duplicated rows
Unique appointments	Checked the number of unique appointment IDs
Outcome categories	Reviewed the values and frequency of appointment outcomes

Data-quality check	Method used
Categorical consistency	Reviewed the unique values in relevant categorical fields
Date fields	Checked and converted date fields for consistency
Booking lead time	Compared booking and appointment dates with recorded lead time
Age groups	Checked the relationship between age and age group
Previous appointment history	Checked the relationship between previous appointments and previous no-shows
Reminder information	Compared reminder status with reminder channel

This approach provides a repeatable method for identifying potential issues before moving into detailed exploratory analysis.

Missing Values

The missing-value check identified missing data in three variables:

Variable	Missing records
reminder_channel	1,366
distance_to_clinic_km	90
waiting_time_minutes	60

The missing values in reminder_channel require particular attention.

On further inspection, the missing reminder-channel values occur where **no reminder was sent**. This means the missing values appear to have a logical explanation rather than automatically indicating poor data entry.

For distance_to_clinic_km and waiting_time_minutes, the number of missing records is much smaller. These variables will need to be considered carefully when we perform the detailed analysis in the next stage.

I would not remove these records automatically at this point. The appropriate treatment will depend on the analysis being performed.

Duplicate Records

The duplicate check identified no duplicate rows.

There are also **5,000 unique appointment IDs across the 5,000 records**.

This is a positive data-quality finding because each appointment appears to be represented by a unique appointment record.

Appointment Outcomes

The dataset contains three appointment outcome categories:

- **Attended**
- **No-Show**
- **Cancelled**

The initial review showed that **No-Show is the largest outcome category**, followed by Attended and Cancelled.

This makes appointment attendance and no-show behaviour an important area for further investigation.

However, at this stage I am treating this as an initial observation rather than a final business conclusion. The Week 4 instructions specifically state that KPIs do not need to be calculated or visualised yet.

Consistency Checks

A number of logical checks were also performed.

The recorded booking_lead_days was compared with the difference between the booking date and appointment date, and the values were consistent.

The age and age_group fields were also checked for consistency.

The previous appointment history was reviewed to ensure that previous_no_shows did not exceed previous_appointments.

The reminder fields were also checked. Records where no reminder was sent did not contain a reminder channel, which explains the missing reminder_channel values.

These checks provide reasonable confidence that the dataset is suitable for the initial exploratory analysis, while still leaving some areas that will require attention during data preparation.

5. Important Variables for the Analysis

After reviewing the dataset and data dictionary, I identified the following variables as particularly relevant.

Appointment outcome

appointment_outcome

This is the main variable for the analysis because it tells us whether an appointment was attended, missed or cancelled.

Previous no-shows

previous_no_shows

This is important because previous attendance behaviour may help us understand patterns in future appointment outcomes.

Previous appointments

previous_appointments

This provides additional context around a patient's previous appointment history.

Reminder information

reminder_sent and reminder_channel

These variables will allow us to investigate whether appointment outcomes differ depending on whether patients received reminders and, where applicable, which channel was used.

Booking lead time

booking_lead_days

This can help us examine whether the amount of time between booking and the appointment is associated with attendance.

Appointment characteristics

appointment_type, appointment_day, and appointment_time

These variables can help identify whether certain appointment types, days or times have different attendance patterns.

Patient characteristics

age, age_group, and gender

These variables can support descriptive comparisons between different patient groups.

Operational factors

distance_to_clinic_km and waiting_time_minutes

These variables may provide additional context around accessibility and the appointment experience.

6. Business Questions

Based on the initial review, I would investigate the following questions.

1. How common are no-shows compared with attended and cancelled appointments?

This will establish the overall appointment outcome pattern.

2. Is previous no-show history associated with future appointment outcomes?

This will help determine whether patients with a history of missed appointments show different attendance patterns.

3. Is appointment attendance associated with reminder activity?

This will examine whether attendance patterns differ between appointments where reminders were sent and those where they were not.

4. Does booking lead time differ between attended, no-show and cancelled appointments?

This could reveal whether the amount of time between booking and appointment is associated with different outcomes.

5. Do appointment outcomes vary by appointment type, day or time?

This could help identify operational patterns that may be useful for clinic scheduling.

6. Are there differences in appointment outcomes across age groups or gender?

This will provide descriptive demographic insight while avoiding unsupported assumptions about patient behaviour.

7. Is there a relationship between distance to the clinic, waiting time and appointment outcomes?

This may help identify whether accessibility or operational factors are worth exploring further.

7. Proposed KPIs

The Week 4 assignment requires **3–5 potential KPIs**, with each KPI connected to a business question. At this stage, the KPIs only need to be identified and justified; they do not need to be calculated or visualised.

I selected five potential KPIs.

KPI 1: No-Show Rate

Business question:
How common are no-shows?

Why it matters:
This would provide a direct measure of the clinic's missed-appointment problem and could be monitored across different periods or appointment groups.

KPI 2: Attendance Rate

Business question:
How does successful attendance compare with other appointment outcomes?

Why it matters:
Attendance rate provides a clear measure of how successfully scheduled appointments are being attended.

KPI 3: Reminder-Associated Attendance Rate

Business question:
Is appointment attendance associated with reminder activity?

Why it matters:
This could help the clinic understand whether reminder activity is associated with better attendance patterns.

KPI 4: Repeat No-Show Rate

Business question:
Is previous no-show behaviour associated with future appointment outcomes?

Why it matters:
This could help identify whether repeated missed appointments are a significant pattern within the dataset.

KPI 5: No-Show Rate by Booking Lead-Time Group

Business question:
Does booking lead time appear to be associated with no-shows?

Why it matters:
This could help identify whether appointments booked further in advance have different attendance patterns.

8. Initial Analysis Approach

The analysis will be carried out in stages rather than jumping directly into dashboard creation.

Stage 1 – Data Preparation

The first stage will involve preparing the data for analysis.

This will include:

- Reviewing data types.
- Confirming date formats.
- Reviewing missing values.
- Checking categorical values.
- Deciding how relevant missing values should be handled.
- Creating any necessary derived fields.
- Keeping the original dataset unchanged.

The assignment specifically instructs interns not to overwrite original project resources and to save cleaned or transformed data separately.

Stage 2 – Exploratory Analysis

The next step will be to explore appointment outcomes using descriptive statistics and appropriate visualisations.

The analysis will examine:

- Overall appointment outcomes.

- No-show patterns.
- Previous no-show history.
- Reminder status and channels.
- Booking lead time.
- Appointment type.
- Appointment day and time.
- Age groups and gender.
- Distance to clinic.
- Waiting time.

Stage 3 – Comparison and Pattern Identification

I will compare appointment outcomes across the key variables to identify meaningful differences.

For example:

- No-shows among patients with and without previous no-shows.
- Outcomes for appointments with and without reminders.
- No-shows across booking lead-time groups.
- No-shows across appointment types.
- Attendance across different appointment days and times.

The aim will be to identify **patterns and relationships**, rather than assume that one variable causes another.

Stage 4 – KPI Analysis

Once the data has been prepared and explored, the selected KPIs will be calculated and analysed.

Stage 5 – Visualisation and Reporting

For the later stages, **Power BI** can be used to communicate the findings through charts, tables and dashboard visuals.

Python/Pandas will remain useful for data preparation and exploratory analysis, while Power BI can be used to present the results in a business-friendly format.

The assignment lists Python, SQL, Power BI and Tableau as appropriate Data Analytics tools, but does not require all of them to be used.

9. Assumptions, Limitations, Risks and Dependencies

Assumptions

I have assumed that the appointment outcome represents the final status of each appointment and that the definitions provided in the data dictionary accurately describe the variables.

I have also treated the missing reminder channel values as potentially meaningful because they occur where no reminder was sent.

Limitations

The dataset is fictional and anonymised.

There are missing values in distance to clinic and waiting time.

The dataset may not contain every factor that influences whether a patient attends an appointment.

Most importantly, an observed relationship between two variables should not automatically be interpreted as a causal relationship.

Risks

Poor handling of missing data could affect the results.

Some groups may contain considerably more records than others, so comparisons will need to take group size into account.

Cancelled appointments should not automatically be treated as no-shows because they are separate outcome categories in the dataset.

Demographic differences should also be interpreted carefully and should not be used to make unsupported assumptions about individual patients.

Dependencies

The analysis depends on:

- Accurate interpretation of the data dictionary.
- The quality of the appointment dataset.
- Appropriate treatment of missing data.

- Consistent definitions for the KPIs.
- Appropriate analytical and visualisation tools.

10. Key Findings from Week 4

The initial review provides a good foundation for the next phase of the project.

The dataset contains **5,000 appointment records across 18 variables**, providing information about patient characteristics, appointment details, previous appointment history, reminders and operational factors.

The data-quality assessment found **no duplicate records**, while missing information was concentrated in three variables: reminder channel, distance to clinic and waiting time.

One useful observation is that the missing reminder-channel values appear to correspond to appointments where no reminder was sent. This suggests that the missingness has a logical explanation and should not automatically be treated as an error.

The review also identified several variables that are potentially useful for understanding appointment outcomes, particularly previous no-shows, reminder activity, booking lead time, appointment characteristics and operational factors.

These findings give us a clear direction for the exploratory analysis that follows.

11. Week 5 Proposed Focus

The main focus for Week 5 will be to move from **initial assessment into actual exploratory data analysis**.

The planned work will include:

1. Preparing the dataset for analysis.
2. Addressing relevant data-quality issues.
3. Calculating the selected KPIs.
4. Exploring attendance and no-show patterns.
5. Comparing outcomes across key variables.
6. Creating appropriate charts and tables.

7. Identifying significant trends and anomalies.
8. Translating the findings into clear business insights.

This follows the assignment's intended progression from **problem understanding and initial analysis into analysis and solution development**.

12. Week 4 Project Summary

Problem addressed

The Data Analytics track is focused on understanding appointment attendance and no-show patterns within the HealthConnect appointment dataset.

Resources used

I reviewed the HealthConnect Appointment Dataset, the HealthConnect Data Dictionary and the Week 4 project assignment requirements.

Key observations

The dataset contains 5,000 records and 18 variables. No duplicate records were identified. Missing values were found in reminder channel, distance to clinic and waiting time. The reminder-channel missing values appear to be related to appointments where no reminder was sent.

Method

The initial data-quality assessment was performed using **Python and Pandas**, with checks covering dataset structure, data types, missing values, duplicates, unique identifiers, categorical values, dates and logical relationships between variables.

Proposed approach

The next phase will use exploratory data analysis to investigate appointment outcomes across previous no-show history, reminders, booking lead time, appointment characteristics, demographics and operational factors.

Key considerations

Missing values will need to be handled appropriately, cancelled appointments should remain distinct from no-shows, and relationships identified in the data should be treated as associations rather than proof of causation.

Week 5 focus

Week 5 will focus on data preparation, KPI calculation, exploratory analysis, visualisation and identifying actionable business insights.